HyTray Invention Disclosure Series 1

Paper III — Helical-Indexed Variable-Slot Swirl Contact Tray (VORTEXA): A Load-Tracking Extension of the Centrifugal Vapour-Swirl Stage

Inventor: Pratik

Invention Disclosure / Provisional Specification Draft — HyTray Series HT-03
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Abstract
A vapour–liquid contacting tray is disclosed in which the swirl-tube architecture of HT-02 (CVS) is augmented by a coaxial helical-indexed sleeve (54) that varies the effective tangential vapour-inlet slot area as a function of axial lift. The sleeve is constrained against fixed stator baffles (60) by a double-flank back-drivable helical path (62), so axial vapour pressure differential is kinematically coupled to rotation: rising vapour load lifts and rotates the sleeve, progressively uncovering additional slot area; falling vapour with rising liquid drives the sleeve down and into a closed condition. Effective slot area thus tracks load passively, with no actuator. Operating turndown is extended from the $\sim$3:1 of fixed-geometry swirl-tube stages to a predicted $\sim$8–10:1, while the centrifugal capacity, pressure drop and spacing-decoupling characteristics of the parent swirl-tube class are preserved. Predicted capacity factor at flood is $C_{sb} = 0.14$–0.18 m/s at tray spacings down to 300 mm. The disclosed inventive matter resides in the helical-indexed slot-modulation sleeve, the double-flank back-drivable kinematic constraint, and the passive load-tracking behaviour these together produce in combination with a swirl-tube contactor — not in the bare concept of swirl contacting, nor in deck-pool liquid metering taken alone (HT-02), which are prior art relative to this disclosure.

Status note: performance values are model predictions (effective slot area derived from helical kinematics; capacity inherited from HT-02 swirl-separation analysis; turndown predicted from quasi-static lift-balance and weep onset). They require CFD and cold-rig validation. Helical-actuated valve hardware is dense prior art in unrelated fields (pressure-relief, surge protection); a professional search distinguishing application to swirl-tube vapour-inlet modulation is mandatory before filing.

1 Field of the Invention

Vapour–liquid contacting internals for distillation and absorption columns operating at or beyond the gravity-settling capacity limit, particularly capacity revamps within existing shells requiring wide operating turndown beyond that achievable with fixed-geometry swirl-tube stages.

2 Background and Prior Art

The capacity of conventional cross-flow trays is bounded by the Souders–Brown criterion. High-capacity decks (SUPERFRAC, VGPlus, UFM and equivalents) recover area from downcomers and remain gravity-limited at $C_{sb} \approx 0.12$ m/s.

Swirl-tube stages (Shell ConSep class, and the CVS disclosure HT-02 of this series) demonstrate centrifugal de-entrainment exceeding the gravity ceiling, reaching $C_{sb} = 0.14$–0.18 m/s by establishing a 30–60 g wall acceleration inside short axial contactors fed tangentially by 4–8 vapour slots and metered with liquid from the deck pool through calibrated ports. The principal residual limitation of such fixed-geometry stages is operating turndown: the slot velocity band ($\sim$18–35 m/s) required to maintain swirl number $S = 1.5$–3.0 admits only a $\sim$3:1 vapour load swing without falling out of the design separation regime; a dual-slot-row variant extends this to $\sim$5:1 at the cost of hardware complication.

The present invention addresses this turndown limitation by varying slot open area in proportion to lift, the lift being driven passively by the vapour-side pressure differential through a helical kinematic constraint. Capacity, pressure drop and spacing-decoupling properties of the swirl-tube parent are otherwise preserved.

3 Objects of the Invention

1. To provide a swirl-tube contacting stage whose effective tangential vapour-inlet area tracks vapour load passively, without external actuation, extending operating turndown beyond the fixed-geometry limit.
2. To maintain swirl number S within the 1.5–3.0 band, and wall acceleration above 30 g, across an instantaneous turndown of at least 8:1 at the cell level.
3. To preserve the $C_{sb} = 0.14$–0.18 m/s capacity, the 8–14 mbar tray pressure drop, and the spacing-decoupling characteristics of the parent swirl-tube architecture.
4. To couple axial sleeve lift to rotation by a kinematically locked helical constraint, such that one degree of freedom modulates both effective slot area and the swirl-injection geometry.
5. To render the constraint back-drivable in both senses (up under vapour push, down under weight and liquid), so that the sleeve passively settles at the lift dictated by the instantaneous balance of vapour and liquid loads.
6. To present a single defensible inventive step — the helical-indexed slot-modulation sleeve — that may be claimed independently of, and combined with, prior-art swirl-tube and deck-pool feed elements.

4 Summary of the Invention

Each contactor cell comprises a vertical swirl tube (40) of 80–150 mm inside diameter and height 1.2–2.0 diameters, penetrating the deck and provided with $N_s = 6$–12 fixed tangential vapour-inlet slots (42) in its upper wall and 3–6 calibrated liquid-metering ports (46) of ∅ 8–14 mm opening from the surrounding deck pool into the tube wall below the slots. A coaxial sleeve (54) surrounds the upper tube and is provided with apertures (56) of width and pitch nominally matching the tube slots. The sleeve is constrained by guide pins or vanes (58) bearing against fixed vertical stator baffles (60) stamped on or attached to the deck; the bearing surfaces of (58) form a double-flank helical profile (62) of lead $L = 150$–260 mm/rev and lead angle $\psi = 25°$–$40°$, with drive and counter flank angles ψ_d, ψ_c both exceeding the friction angle $\phi = \arctan \mu$ so that the path is back-drivable in both senses. Axial lift z of the sleeve and its rotation θ are kinematically locked, $z = L\theta/(2\pi)$. Effective slot open area $A_s(z)$ varies monotonically and continuously from zero (sleeve fully seated) to the geometric maximum $A_{s,\max} = N_s w_s h_s$ (sleeve fully lifted), in proportion to the registration of apertures (56) against slots (42). At the tube exit, a crown separator (48) of 8–16 vanes pitched co-rotationally at 35°–55° intercepts the swirling two-phase flow and discharges separated liquid into a collection annulus (50) draining via sealed conduits directly to the outlet-weir region of the same tray.

5 Brief Description of the Drawings

Figure 1: (a) elevation section of one swirl tube with sleeve in the low-load (seated) condition — effective slot area ≈ 0, sleeve closed against deck; (b) elevation section of the same cell in the high-load (fully lifted) condition — effective slot area $= A_{s,\max}$, apertures registered with slots, full swirl established; (c) plan section showing fixed baffles (60), sleeve guide vanes (58), and the helical engagement.

Figure 2: (a) effective slot area $A_s(z)$ and required tangential velocity $V_\theta(z)$ versus sleeve lift, against the swirl-number target band; (b) predicted instantaneous turndown band of the present cell against the fixed-geometry swirl tube and conventional cross-flow trays.

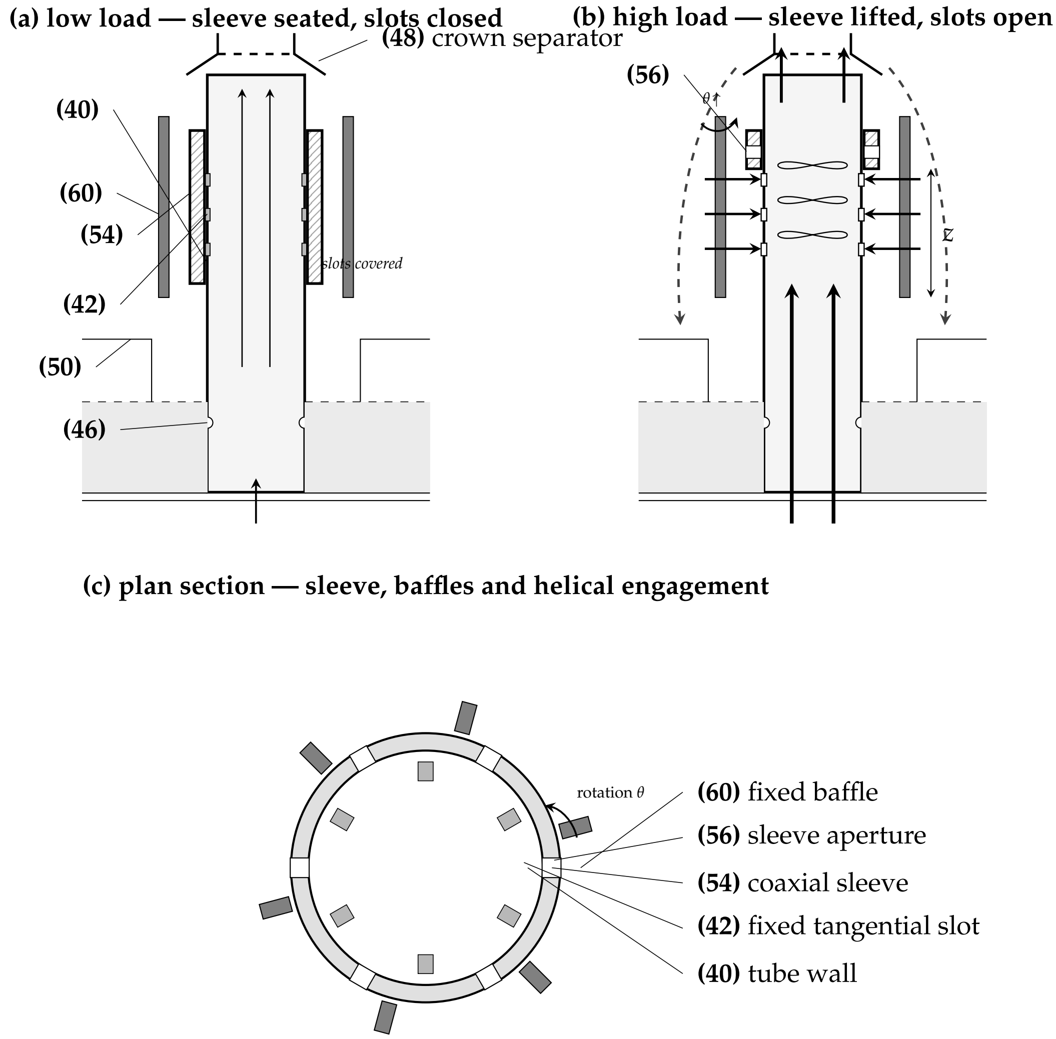


Figure 1. Swirl contactor cell with helical-indexed slot-modulation sleeve. Tube (40), tangential slots (42), liquid ports (46), crown separator (48), collection annulus (50), sleeve (54), sleeve aperture (56), guide vane (58), fixed baffle (60), helical engagement profile (62).

6 Detailed Description

6.1 Helical kinematic constraint

The sleeve (54) is constrained against the fixed baffles (60) along helical engagement surfaces (62). For the preferred tube ID of 110 mm and a sleeve thickness of 6 mm, the helix mean radius is $r_h = 55$–70 mm. The lead angle $\psi = \arctan(L/2\pi r_h)$ is held in the back-drivable range 25°–40°, which fixes the helix lead at $L = 2\pi r_h \tan\psi = 150$–$260$ mm/rev (preferred $L \approx 170$ mm/rev giving $\psi \approx 25°$, chosen at the shallow end for finer rotational modulation). Sleeve axial lift z and angular rotation θ are kinematically locked by the constraint:

$$z = \frac{L}{2\pi}\,\theta, \qquad \theta = \frac{2\pi z}{L}.$$

Because the full design lift range is only $z = 0$–20 mm, the corresponding rotation across the entire operating band is:

$$\theta_{\max} = \frac{2\pi z_{\max}}{L} = \frac{360^\circ\, z_{\max}}{L} \;\approx\; 28^\circ\text{–}48^\circ.$$

The sleeve therefore executes **less than half a turn end-to-end**; the mechanism is a *partial-turn indexing* device, not a fully rotating screw. Apertures (56) and slots (42) are pitched so that this $\sim$30–50° excursion takes each aperture from full coverage of its associated slot to full registration.

The engagement profile is double-flanked: the drive flank (lead angle ψ_d, contacted under vapour push) and the counter flank (relief angle ψ_c, contacted under sleeve weight and liquid load). Both flank angles exceed the friction angle $\phi = \arctan \mu$ of the bearing surfaces ($\mu \approx$ 0.10–0.15 for rolling guides), so the constraint is back-drivable in both senses:

$$\tan \psi_d > \mu \quad \text{and} \quad \tan \psi_c > \mu.$$

Setting $\psi_d \neq \psi_c$ tunes the lift and re-seat thresholds independently and provides an anti-chatter hysteresis band of width approximately $(\tan \psi_d - \tan \psi_c) \cdot \mu \cdot W_{\text{sleeve}} / A_{\text{sleeve}}$ at the transition.

Performance sensitivity to helix parameters:

- *Lead angle ψ.* Shallower helix (closer to $\arctan \mu$) gives more lift per unit pressure differential — the sleeve responds faster and onset occurs at lower vapour load, at the cost of reduced stability margin against back-driving by friction alone. Steeper helix gives slower, more stable response with a higher lift-onset threshold. The preferred 30° centres the design at $\sim 2\times$ the friction angle.
- *Helix lead L.* Sets the rotation rate per unit lift, $d\theta/dz = 2\pi/L$. Larger L (steeper helix) gives less rotation per mm lift, requiring wider apertures relative to slot pitch; smaller L (shallower) gives more rotation, allowing narrower apertures and finer modulation.
- *Number of turns.* Held below half a turn deliberately — it bounds the angular travel of bearing surfaces, limits cumulative friction over the operating cycle, and confines wear to a small arc of the engagement. Trying to use multiple turns of a longer helix is mechanically inferior: more arc, more wear, more chance of binding.
- *Mean radius r_h.* Larger r_h at fixed ψ requires longer L but gives more leverage of vapour-pressure-induced torque against friction — generally improves back-drivability. Tight against the tube ($r_h \approx$ tube OD/2) is mechanically convenient but reduces leverage.

6.2 Effective slot area as a function of lift

Apertures (56) in the sleeve are of nominal height h_a and width w_a at pitch matching the tube slots (42). Effective slot open area is the overlap integral:

$$A_s(z) = N_s\, w_s\, h_{s,\text{open}}(z)\, \phi_\theta\big(z/L_z\big),$$

where $h_{s,\text{open}}(z) = \min(h_s,\ \max(0,\ z + h_s - h_{cl,s}))$ is the vertical overlap, $h_{cl,s}$ the sleeve–slot overlap at $z = 0$, and $\phi_\theta \in [0, 1]$ is the rotational registration factor set by the helix. With $h_{cl,s} = h_s$ (sleeve fully covers slots when seated) and ϕ_θ tuned to reach unity at $z = z_{\max}$, $A_s(z)$ is a monotonic continuous function from 0 at $z = 0$ to $A_{s,\max} = N_s w_s h_s$ at $z = z_{\max} \approx h_s$. The cell thus behaves as a continuously variable nozzle ring.

6.3 Slot geometry and tangential guide vanes

The tangential vapour-inlet slots (42) are openings in the wall of the fixed tube (40), formed by short **curved guide vanes (43)** welded to the leading and trailing edges of each opening. The vanes set the inclination angle β of the exit jet and ensure clean tangential injection — without them, vapour expanding through a plain rectangular opening would substantially straighten and lose much of its tangential momentum. The vanes are not on the moving sleeve; the sleeve apertures (56) only modulate *how much* of each vane-defined slot is exposed.

Preferred slot dimensions (per tube, for 110 mm tube ID):

Slot dimension	Claimed band	Preferred
Slot height h_s	25–40 mm	30 mm
Slot width w_s	5–12 mm	8 mm
Slot count N_s	6–12	8
Slot inclination β	10°–30°	20°
Guide vane height above tube ID	4–8 mm	6 mm
Vane leading-edge radius	1–2 mm	1.5 mm
Vane curvature (turning angle)	$\beta + (5\text{–}10^\circ)$	$\beta + 8^\circ$
Total slot open area $A_{s,\max} = N_s w_s h_s$	900–3000 mm^2	1920 mm^2

Vane turning angle is biased slightly above β (an "overturn" of 5–10°) because viscous straightening on the suction side of the vane recovers a few degrees toward axial; the residual exit angle after this recovery matches the swirl-number target $S = \tan\beta$.

6.4 Vapour path and swirl generation

Slot exit velocity is held within 18–35 m/s across the operating window by the lift-modulated area:

$$v_s = \frac{\dot{Q}_V}{C_d\, A_s(z)}, \qquad V_\theta = v_s \sin\beta, \qquad V_{ax,s} = v_s \cos\beta,$$

with $C_d \approx 0.7$. Slot inclination $\beta = 10°\text{–}30°$ from the tangent fixes the partition between tangential and axial components. Swirl number $S = V_\theta / V_{ax} \approx \tan\beta$ is held in the 1.5–3.0 band; wall acceleration $a_c = V_\theta^2/r$ reaches 30–60 g at design lift, yielding capacity factor $C_{sb,\text{swirl}} = C_{sb,\text{grav}}\sqrt{a_c/g}$, eq. (1).

$$C_{sb,\text{swirl}} \;=\; C_{sb,\text{grav}}\sqrt{a_c/g} \tag{1}$$

The principal contribution of the sleeve mechanism in this regime is that v_s is held within the design band even as $\dot{Q}_V$ varies across turndown: $A_s(z)$ falls as $\dot{Q}_V$ falls, so the slot-velocity ratio remains close to its design value. A fixed-area swirl tube (HT-02 class) loses swirl number below ~30% load because v_s falls proportionally; the HT-03 sleeve preserves $S \geq 1.5$ across the full operating band, which in turn preserves the centrifugal separation field throughout turndown.

6.5 Quasi-static force balance setting the lift

The sleeve floats at the z where vapour-side lift balances weight, liquid loading and friction:

$$\left[\Delta P_{\text{riser}} + \Delta P_{\text{rev}} + \Delta P_{\text{slot}}(z)\right] A_{\text{sleeve}} \;+\; \dot{m}_V V_\theta\, r_h \sin\psi \;=\; W_{\text{sleeve}} + F_{\text{liq}} \;\pm\; F_\mu.$$

Because the dry pressure-drop terms scale as v_s^2 and $v_s \propto 1/A_s(z)$, the lift function is self-stabilising: a small positive perturbation in z raises A_s, drops v_s and ΔP_{slot}, reducing the lift force — the sleeve settles to the equilibrium z rather than running to the end-stop. Sleeve weight is selected so that lift onset occurs at $u_r \approx 6\text{–}8$ m/s riser velocity (well below design 25 m/s), giving a usable lift band $z = 2\text{–}20$ mm over the operating range.

6.6 Liquid metering from the deck pool

Liquid is metered from the surrounding deck pool through 3–6 calibrated submerged orifice ports (46) of ∅ 8–14 mm. Per-port flow follows the standard submerged orifice law $Q_p = C_d A_p \sqrt{2gh_{cl}}$ with $C_d \approx 0.62$; per-tube load self-equalises across the deck because all ports share a single pool. Ports are located below the lowest slot elevation, so liquid is sheared into the rotating film rather than falling through the vortex core. Liquid metering is independent of sleeve position — the sleeve modulates only the vapour-inlet area, not the liquid feed — so the L/V partition tracks the column duty automatically across the turndown range.

6.7 Annular film regime

The film Reynolds number $\mathrm{Re}_f = 4\Gamma/\mu_L$ at the inner tube wall is held in the 1 500–8 000 band by the combination of port sizing and pool depth; the band is robust to sleeve lift because the liquid path is decoupled from the sleeve.

6.8 Crown separation and re-entrainment control

Crown vanes (48) at the tube exit turn the swirling film into the collection annulus (50). The annulus drains via sealed conduits to the outlet-weir region of the same tray, bypassing active bubbling area entirely; this is the parent-disclosure (HT-02) liquid-return path, unchanged. Sealing legs prevent vapour bypass.

6.9 Layout and hydraulics

Tubes occupy 25–40% of column area on a triangular pitch of 1.4–1.7 tube diameters, as for HT-02. Tray pressure drop is the sum of riser, reversal, lifted-slot and crown contributions:

$$\Delta P_t = \underbrace{(K_r + K_{\mathrm{rev}})\tfrac{1}{2}\rho_V u_r^2}_{\text{tube path}} + \underbrace{K_s(\beta)\tfrac{1}{2}\rho_V v_s^2(z)}_{\text{lift-modulated slot}} + \Delta P_{\mathrm{crown}},$$

predicted 9–14 mbar across the operating window. Because v_s is held within a narrow band by the lift mechanism, ΔP_t remains nearly load-invariant — a beneficial property for column control.

6.10 Turndown

The combination of (i) lift-modulated slot area and (ii) the bubble-cap-equivalent static seal at $z = 0$ (sleeve seated below the deck pool surface, equivalent to slot submergence) gives a single-cell instantaneous turndown of $\sim$8:1, with a corresponding swirl-number maintenance of $S \geq 1.5$ across the band. Cell-staging across the deck (a fraction of cells held closed by heavier sleeves engaging only at higher load) extends deck-level turndown into the 10:1 region; this staging is implemented by sleeve weight selection alone, with no mechanical interlock.

6.11 Swirl persistence and centrifugal disengagement

Vapour entering through slots (42) at velocity v_s with $V_\theta = v_s \sin\beta$ establishes a Rankine-type vortex in the upper tube. Angular momentum decay along the tube is governed by wall and inter-phase shear; for tubes of $H/D = 1.2$–2.0 the swirl number at the crown exit retains **70–85%** of its inlet value. Within this rotating field, droplets entrained at the inlet experience radial migration at

$$u_r = \frac{d^2(\rho_L - \rho_V)\, a_c}{18\,\mu_V}, \qquad a_c = \frac{V_\theta^2}{r},$$

yielding a Lapple-type cut diameter

$$d_{50} \;\propto\; \sqrt{\frac{18\,\mu_V\, R}{a_c\, \Delta\rho\, t_{\mathrm{res}}}}.$$

At 45 g, 110 mm tube, and design residence time $t_{\mathrm{res}} \approx 36$ ms, $d_{50} \approx 25$–30 μm — comfortably below the droplet sizes that drive entrainment flooding. The radial migration time across the half-tube ($\sim$55 mm) for a 50 μm droplet is approximately 11 ms, well inside the residence time, so the entire droplet cloud is delivered to the wall film before reaching the crown.

The crown vanes (48) at the tube exit do two things simultaneously: they catch the spinning annular film and discharge it tangentially into the collection annulus (50), and they recover residual rotational kinetic energy from the remaining vapour by partially de-swirling it before it reaches the tray above. The combination of in-tube centrifugal capture plus crown collection yields predicted entrainment ratio $\psi < 0.02$ at 90% flood, equal to HT-02 fixed-geometry performance.

Critically, the sleeve mechanism preserves this separation across turndown: because $A_s(z)$ tracks $\dot{Q}_V$, the slot velocity v_s and therefore V_θ remain in the design band even at low load, so the swirl number S does not collapse and the centrifugal field does not decay below $\sim$30 g. Fixed-geometry swirl-tube stages lose this field below $\sim$30% load and revert to gravity-only disengagement; the HT-03 sleeve specifically prevents this regime loss.

6.12 Fouling, reliability and failure modes

Mechanical reliability of the sleeve–baffle engagement is the principal durability question for HT-03. The following design choices mitigate the dominant failure modes:

- **Rolling rather than sliding contact.** Sleeve guide vanes (58) bear on baffle (60) faces through small rolling contacts (or low-friction inserts), not flat sliding pads. Foulant cannot wedge between two converging rolling surfaces as it can between sliding pads.
- **Self-sweeping motion.** The sleeve traverses its 28–48° arc many times daily under load variations, each excursion bringing a fresh arc of bearing surface into contact and displacing any accumulated solids.
- **Working clearance.** Sleeve-to-baffle clearance is held at 0.5–1.0 mm (preferred 0.7 mm); particles smaller than this pass through the engagement without contact, particles larger are excluded by the upstream port screens.
- **Drained pathways.** The sleeve top and bottom are open to the deck pool and the annulus respectively, so no stagnant liquid pocket exists where deposits could form by evaporation or settling.
- **Material selection.** Stainless 316L electropolished for general service; duplex 2205 or 2507 for chloride-bearing duty; PTFE-impregnated coatings on guide-vane bearing faces in salt or polymer service.

Failure mode is benign. If the sleeve binds at any intermediate position, the cell does not crash — it reverts to fixed-area swirl-tube behaviour at whatever effective slot area the sleeve was stuck at. A cell jammed in the open position contributes capacity but loses anti-weep at low load; a cell jammed in a partially seated position behaves like a fixed-area swirl tube at that opening. The column continues to operate within HT-02-class performance with reduced turndown extension. There is no scenario in which a stuck sleeve obstructs the riser or blocks the liquid ports.

Service restriction. HT-03 is not recommended for heavy-fouling duty (slop-cuts, asphaltic streams, polymerising or coking services, slurries), where deposit growth on the sleeve guides or in the working clearance is likely to exceed the self-sweeping capacity. For such services the fixed-geometry HT-02 (CVS) without moving parts is the preferred option; HT-03 is targeted at clean and moderate-fouling service where the turndown extension justifies the additional mechanical complexity.

7 Parametric Basis

This section consolidates the governing relations for slot flow, capacity, the helix–capacity coupling, efficiency, and the baffle constraint. All relations are model-level; constants (C_d, K, f, a_i, t_c) require CFD and cold-rig calibration before use in rating.

7.1 Slot flow physics (the cut and the jet)

A slot of width w_s, height h_s, inclination β, with N_s slots and discharge coefficient C_d, behaves as a tangential nozzle. The effective area at sleeve lift z is the aperture–slot overlap:

$$A_s(z) = N_s\, w_s\, h_{s,\text{open}}(z), \qquad h_{s,\text{open}}(z) = \min\big(h_s,\ \max(0,\ z + h_s - h_{cl,s})\big).$$

Slot (jet) velocity, its tangential/axial split, and the slot pressure drop:

$$v_s = \frac{\dot{Q}_V}{C_d\, A_s(z)}, \qquad V_\theta = v_s \sin\beta, \quad V_{ax} = v_s \cos\beta, \qquad \Delta P_{\text{slot}} = \frac{\rho_V v_s^2}{2\, C_d^2}.$$

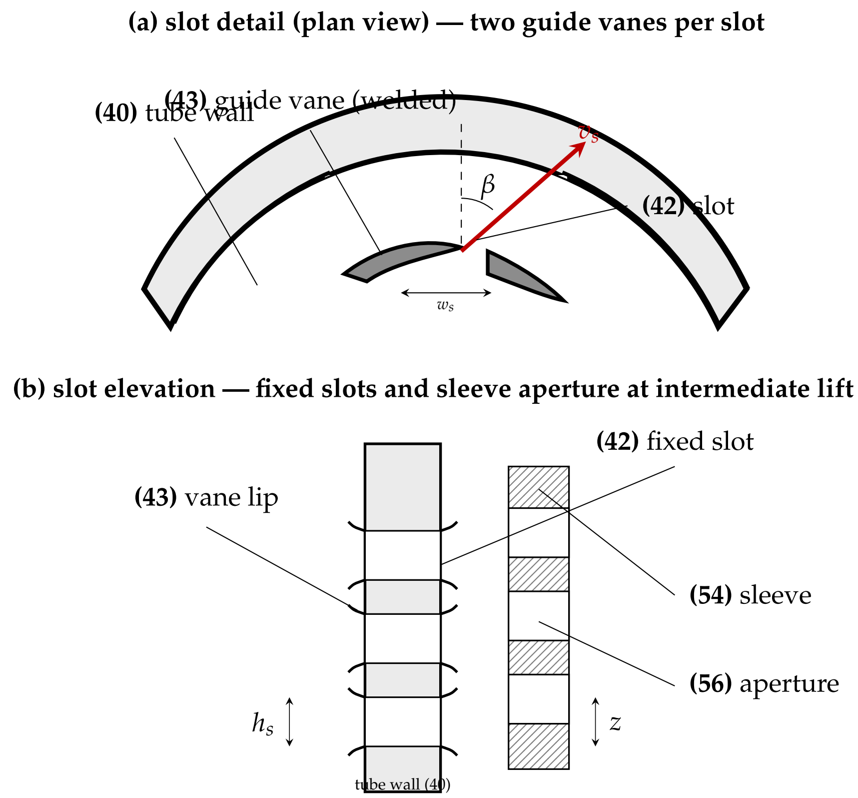


Figure 2. Slot geometry. (a) Plan view of one tangential slot (42) formed in the tube wall (40) by two curved guide vanes (43) welded to the leading and trailing edges of the opening, setting jet inclination β and ensuring tangential exit direction v_s. (b) Elevation showing the column of three slot openings of height h_s in the fixed tube wall, the welded vane lips at the leading edges, and the partially registered sleeve apertures (56) at intermediate lift z. The guide vanes (43) are fixed on the tube — they do not translate or rotate with the sleeve.

Swirl number (ratio of angular to axial momentum flux), for thin tangential slots at radius R:

$$S = \frac{\dot{G}_\theta}{R\,\dot{G}_x} = \frac{\int \rho V_\theta V_{ax}\, r^2\, dr}{R \int \rho V_{ax}^2\, r\, dr} \approx \tan\beta.$$

The design band $S = 1.5$–3.0 avoids vortex breakdown reaching the liquid ports; slot aspect ratio $h_s/w_s \approx 3$–5 keeps each jet coherent and avoids early merging that would reduce wall coverage.

7.2 Centrifugal field and capacity

Wall acceleration and the capacity factor (Souders–Brown generalised with body force a_c):

$$a_c = \frac{V_\theta^2}{r_h}, \qquad N_g = \frac{a_c}{g}, \qquad \boxed{C_{sb,\mathrm{swirl}} = C_{sb,\mathrm{grav}}\sqrt{N_g} = C_{sb,\mathrm{grav}}\,\frac{V_\theta}{\sqrt{r_h\, g}}}.$$

Droplet cut size (Lapple/Stokes; vortex of acceleration a_c, radial travel R, residence t) and radial migration:

$$d_{50} \propto \sqrt{\frac{18\,\mu_V\, R}{a_c\,\Delta\rho\, t}}, \qquad u_r = \frac{d^2 \Delta\rho\, a_c}{18\,\mu_V}, \qquad t_{\mathrm{res}}\, u_r \geq \Delta r.$$

7.3 How capacity varies with helix rotation

The helix locks lift to rotation, $z = L\theta/2\pi$. To preserve the swirl field, the passive control law holds v_s near its design value v_s^* by opening area as flow rises; throughput at constant slot velocity is then set by open area, which is set by lift, which is set by rotation:

$$\dot{Q}_V = C_d\,A_s(z)\,v_s^* = C_d\,v_s^*\,N_s w_s\,h_{s,\mathrm{open}}\!\left(\tfrac{L\theta}{2\pi}\right).$$

In the linear registration band ($h_{s,\mathrm{open}} \propto z \propto \theta$):

$$\boxed{\dot{Q}_V(\theta)\ \approx\ \dot{Q}_{V,\max}\,\frac{\theta-\theta_0}{\theta_{\max}-\theta_0}} \qquad \Longrightarrow \qquad \text{throughput is (nearly) linear in helix rotation.}$$

The intensive capacity factor C_{sb} stays *constant* across this band (the field is maintained), while the absolute vapour rate scales with θ. Since $d\dot{Q}_V/d\theta = C_d v_s^* N_s w_s\,(L/2\pi)(dh_{s,\mathrm{open}}/dz)$, a *shallower* helix (smaller L) gives finer throughput resolution per degree of rotation, a *steeper* helix (larger L) gives coarser, faster opening — the design lever connecting the mechanism to capacity.

7.4 Mass-transfer efficiency

Murphree point efficiency from transfer units, gas-film controlling in the high-shear swirling film:

$$E_{OG} = 1 - e^{-N_{OG}}, \qquad N_{OG} = \frac{1}{\dfrac{1}{N_G} + \dfrac{\lambda}{N_L}}, \qquad \lambda = \frac{mV}{L}.$$

Transfer units with interfacial area per unit volume a_i and contact (residence) time; penetration-theory liquid coefficient:

$$N_G = k_G\,a_i\,t_{\mathrm{res}}, \qquad t_{\mathrm{res}} = \frac{H}{V_{ax}}, \qquad k_L = 2\sqrt{\frac{D_L}{\pi\,t_c}}, \quad t_c \sim \frac{\nu_L}{u_\tau^2}.$$

Interfacial area sums the swirling wall film and the stripped droplet cloud; surface renewal scales with friction velocity:

$$a_i = \underbrace{\frac{2}{r_{\mathrm{film}}}}_{\text{annular film}} + \underbrace{\frac{6\,\varepsilon_d}{d_{32}}}_{\text{droplet cloud}}, \qquad u_\tau = \sqrt{\tau_w/\rho_L}, \quad \tau_w \propto \rho_V V_\theta^2 f.$$

Because u_τ rises with V_θ and the sleeve keeps V_θ in band across turndown, E_{OG} is sustained at $\approx$ 0.80–0.92 rather than collapsing at low load.

7.5 Fixed baffles and the kinematic constraint

The N_b fixed baffles react the sleeve torque and enforce the helix. Lift–rotation lock and back-drivability:

$$z = \frac{L}{2\pi}\theta, \qquad \tan\psi = \frac{L}{2\pi r_h}, \qquad \tan\psi_d > \mu \text{ and } \tan\psi_c > \mu.$$

Quasi-static float position (lift where vapour force balances weight, liquid and friction):

$$\left[\Delta P_{\mathrm{riser}} + \Delta P_{\mathrm{rev}} + \Delta P_{\mathrm{slot}}(z)\right] A_{\mathrm{sleeve}} + \dot{m}_V V_\theta r_h \sin\psi = W_{\mathrm{sleeve}} + F_{\mathrm{liq}} \pm F_\mu.$$

Lateral bearing reaction per baffle and contact stress (sizes the baffle face and guide):

$$F_{\mathrm{baffle}} = \frac{T_{\mathrm{helix}}}{N_b\,r_h}, \qquad T_{\mathrm{helix}} = F_{\mathrm{axial}}\,\frac{L}{2\pi}\,\frac{\tan\psi + \mu}{1 - \mu\tan\psi}, \qquad \sigma_{\mathrm{contact}} = \frac{F_{\mathrm{baffle}}}{A_{\mathrm{contact}}}.$$

Self-stabilisation: $\Delta P_{\mathrm{slot}} \propto v_s^2 \propto A_s^{-2}$ decreases in z, so $dF_{\mathrm{lift}}/dz < 0$ near equilibrium — the sleeve returns to its float point rather than running to the end-stop, giving a stable, actuator-free operating position at every load.

8 Design Parameter Envelope (claim-band table)

Parameter	Claimed band	Preferred
Tube ID	80–150 mm	110 mm
Tube height / ID	1.2–2.0	1.6
Tangential slots per tube N_s	6–12	8
Slot exit velocity v_s (at design lift)	18–35 m/s	25 m/s
Slot inclination β	10°–30°	20°
Swirl number S	1.5–3.0	2.2
Wall acceleration a_c	30–60 g	45 g
Sleeve thickness	4–10 mm	6 mm
Sleeve lift range z	0–20 mm	0–15 mm
Helix mean radius r_h	50–75 mm	60 mm
Helix lead L	150–260 mm/rev	170 mm/rev
Helix lead angle ψ	25°–40°	25°
Sleeve rotation across full lift $\theta_{\max}$	28°–48°	43°
Drive flank angle ψ_d	25°–40°	32°
Counter flank angle ψ_c	25°–40°	28°
Bearing friction μ (rolling guide)	0.08–0.18	0.12
Liquid ports per tube	3–6, ∅ 8–14 mm	4 × ∅10
Per-tube liquid load	1.5–4.0 m^3/h	2.5
Film Reynolds $4\Gamma/\mu_L$	1 500–8 000	4 000
Crown vanes / pitch angle	8–16 / 35°–55°	12 / 45°
Tube area fraction	25–40 %	32 %
C_{sb} at flood	0.14–0.18 m/s	0.16
Tray spacing	300–450 mm	350 mm
ΔP per tray	9–14 mbar	11
Operating turndown (cell, instantaneous)	6:1–10:1	8:1

Table 1. Claim-band table; ranges are dependent-claim fallback positions.

9 Predicted Performance

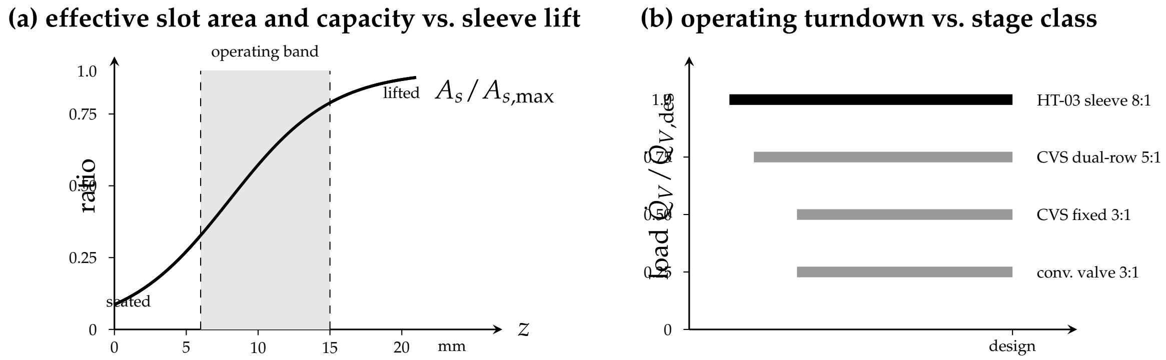


Figure 3. Predicted performance (model-based, pending validation): (a) effective slot area $A_s(z)$ vs. sleeve lift, sigmoidal between seated and lifted limits with the design band shaded; (b) operating turndown band of the present cell against conventional valve, fixed-geometry CVS, and CVS dual-slot-row variant.

Metric	Conv. valve	CVS (fixed)	CVS (dual-row)	HT-03 (predicted)
C_{sb} at flood, 450 mm TS	0.085	0.16	0.15	0.16
Capacity at 300 mm TS (rel.)	0.62	0.97	0.92	0.97
E_{OG} at design	0.72	0.86	0.83	0.85
Entrainment ψ at 90% flood	0.12	<0.02	0.03	<0.02
ΔP / theoretical stage	8 mbar	12 mbar	12 mbar	11 mbar
Operating turndown	3:1	3:1	5:1	8:1–10:1
Cell mechanical complexity	low	low	moderate	moderate (sleeve)

Table 2. Standard hydrocarbon system, SF = 1.0; HT-03 values are model predictions building on the HT-02 swirl-tube performance basis.

10 Claims (draft)

Claim 1. A vapour–liquid contacting tray comprising a deck and a plurality of vertical swirl tubes penetrating said deck, each tube having a plurality of fixed tangential vapour-inlet slots and a plurality of liquid-metering ports opening from a deck pool into the tube wall below said slots, characterised in that each tube is provided with a coaxial sleeve constrained to translate axially and rotate concurrently along a helical path of lead angle ψ, said sleeve having apertures arranged such that the effective open area of said tangential slots is a continuous monotonic function of the axial position of said sleeve.

Claim 2. The tray of claim 1, wherein said helical path is back-drivable in both axial senses, the lead angle ψ exceeding the friction angle $\phi = \arctan \mu$ of the bearing surfaces, such that said sleeve passively settles at an axial position determined by the quasi-static balance of vapour and liquid forces.

Claim 3. The tray of claim 2, wherein the helical engagement comprises a drive flank of angle ψ_d and a counter flank of angle ψ_c, both exceeding ϕ, with $\psi_d \neq \psi_c$ such that the lift and re-seat thresholds of said sleeve are independently set and a hysteresis band suppresses chatter at the transition.

Claim 4. The tray of claim 1, wherein said sleeve is supported and constrained by guide vanes bearing against a plurality of fixed vertical stator baffles formed on or attached to said deck, said baffles further serving to angularly index said sleeve apertures against said slots.

Claim 5. The tray of claim 1, wherein said helical path has a lead L of 150–260 mm/rev and a lead angle ψ of 25°–40°.

Claim 6. The tray of claim 1, wherein each tube has 6–12 fixed tangential slots inclined at 10°–30° from the local tangent, and the sleeve apertures are sized and pitched such that the effective slot area varies between approximately zero and a geometric maximum $N_s w_s h_s$ across an axial lift of 0–20 mm.

Claim 7. The tray of claim 1, wherein the slots and sleeve geometry are such that slot exit velocity is held within 18–35 m/s, swirl number S within 1.5–3.0 and wall centripetal acceleration within 30–60 g across a vapour-load turndown of at least 8:1.

Claim 8. The tray of claim 1, further comprising at the upper end of each tube a crown separator of 8–16 vanes pitched co-rotationally at 35°–55°, discharging separated liquid into a collection annulus, said annulus draining through sealed conduits to the outlet-weir region of the same tray without traversing active bubbling area.

Claim 9. The tray of claim 1, wherein the tubes have an inside diameter of 80–150 mm, a height of 1.2–2.0 diameters, and occupy 25–40% of the column cross-section, and the tray operates at a capacity factor C_{sb} of at least 0.14 m/s at flood at a tray spacing of 450 mm or less.

Claim 10. A method of revamping a distillation column comprising replacing existing cross-flow or fixed-geometry swirl-tube trays with trays according to any preceding claim at a tray spacing of 300–450 mm, thereby increasing column capacity by at least 50% within the existing shell and extending operating turndown to at least 8:1.

Claim 11. A distillation or absorption column containing trays according to any preceding claim.

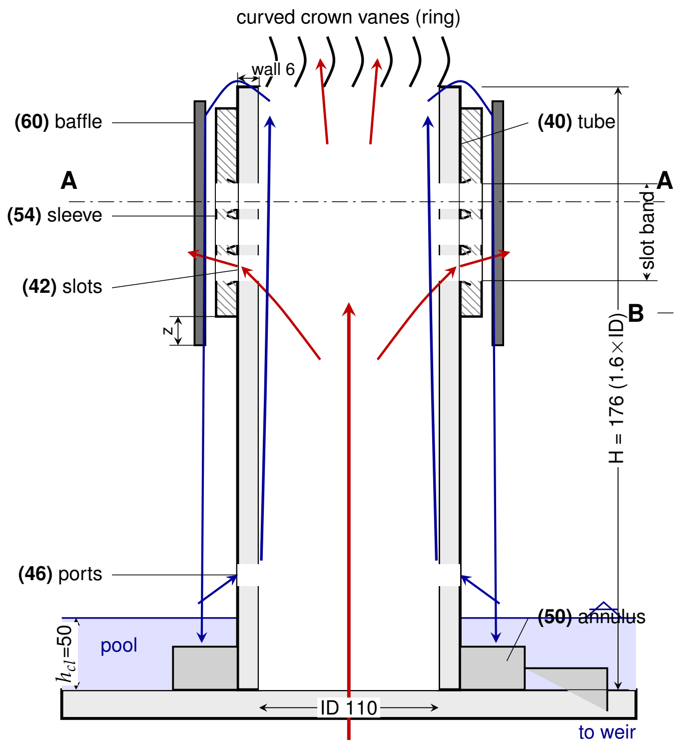
11 Industrial Applicability

Shell-limited revamps in which both capacity and turndown are simultaneously constrained: refinery main fractionators with seasonal feed-rate swings, FCC top sections, demethanisers and C3 splitters subject to upstream variability, offshore columns of expensive footprint where shutdown for load matching is intolerable, and amine/glycol regenerators operating across heavily varying inlet acid-gas loadings. The disclosed sleeve mechanism may also be retrofitted to fixed-geometry swirl-tube trays of the HT-02 class as a turndown-extension upgrade without modification of the tube, port or annulus geometry.

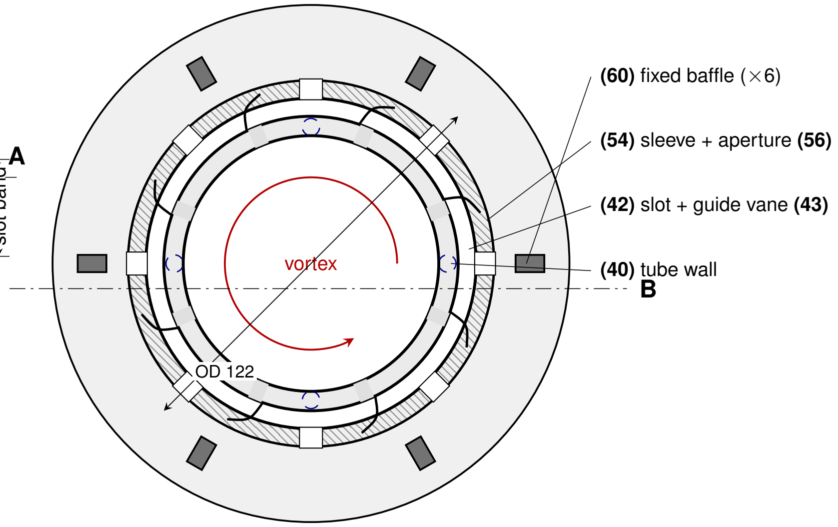
Prior-art neighbours to clear: Shell ConSep/swirl-tube patents and continuations; HyTray Series HT-02 (CVS); helical-actuated and screw-lift valve hardware in pressure relief, fluidic control and turbo-machinery (distinguishable by application to swirl-tube vapour-inlet modulation rather than line throttling); rotary throttle and indexing-sleeve patents in non-contacting equipment. The asserted inventive step is the helical-indexed slot-modulation sleeve in combination with a swirl-tube contactor, the double-flank back-drivable constraint, and the passive load-tracking behaviour.

HT-03 VORTEXA — Construction Drawing Set
(model geometry; all dimensions mm; not certified for fabrication)

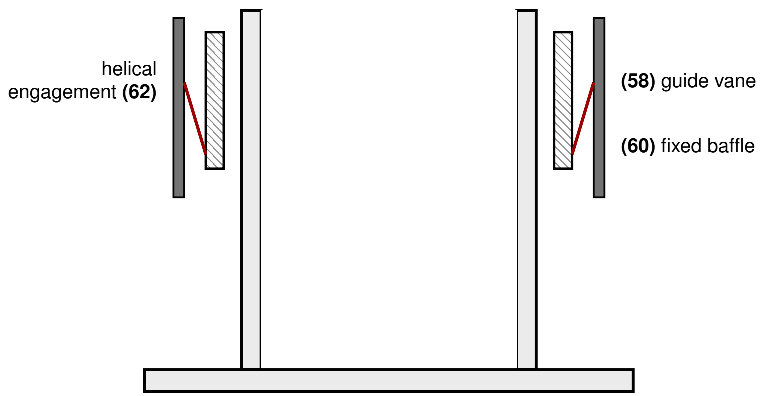
SECTION A–A (front elevation)



PLAN VIEW (looking down, sleeve removed at right half)

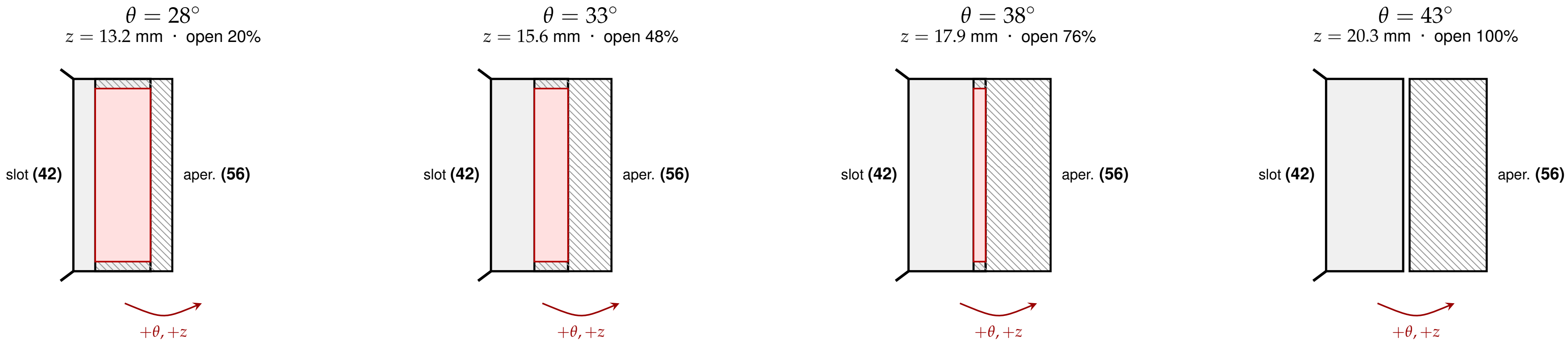


SECTION B–B (side; baffle/guide engagement)



HT-03 VORTEXA — Indexing Sequence & Detail Views

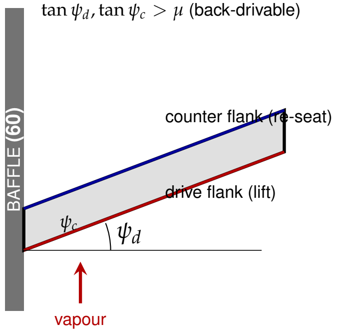
Slot-opening vs sleeve rotation (developed view of one slot; sleeve aperture moves up-and-around the helix; L=170 mm/rev, $z = L\theta/360$):



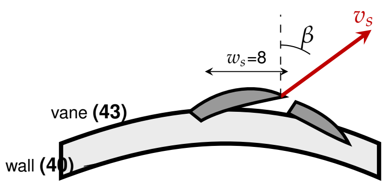
Red = exposed (open) slot area. As the sleeve rotates from 28° to 43° (lift 13→20 mm), the aperture registers progressively with the fixed slot; open area grows ≈20→100%. Onset of opening is below 28°; full registration (end-stop) at 43°.

DETAIL X — helix double flank

$\tan\psi_d, \tan\psi_c > \mu$ (back-drivable)



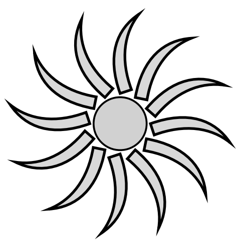
DETAIL Y — slot & guide vane



h_s=30, w_s=8, N_s=8
β=20°, vane turn $= \beta$+8°
$A_{s,\max} = N_s w_s h_s$=1920 mm²

DETAIL Z — crown vane

curved stator ring —
not a flat plate



12 curved vanes @ 45°

(48) crown separator